# A $\varphi$-Harmonic Window for the Alpha Band: Derivation from Cost Geometry, Cube Combinatorics, and Bio-Clocking with Machine-Checked Proofs

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**Abstract**

Within the Recognition Science (RS) framework, we derive a frequency window (7.5, 9.3) Hz from the forcing chain of the Recognition Composition Law. The derivation proceeds in seven steps, each machine-verified in Lean 4 with zero `sorry`: (1) the cost function $J(x) = \frac{1}{2}(x+x^{-1})-1$ is uniquely forced; (2) self-similarity forces $\varphi = (1+\sqrt{5})/2$; (3) $D = 3$ forces the 8-tick period and $\mathrm{lcm}(8, 45) = 360$; (4) a structural hypothesis $N = 360/6 = 60$ selects the neural binding rung; (5) bio-clocking at rung 60 yields $f_{\mathrm{cycle}} \in (38, 45)$ Hz; (6) DFT-8 gives Mode 1 $\in$ (4.7, 5.7) Hz; (7) the unique self-similar ratio $\varphi$ lifts Mode 1 to $f_\varphi \in (7.5, 9.3)$ Hz. Steps 1–3 and 6–7 are theorems; step 4 is a structural hypothesis (face-projection of the sync period); step 5 depends on a single SI calibration $\tau_0 = 7.30 \times 10^{-15}$ s. The derived window contains the Schumann resonance (7.83 Hz), the human alpha rhythm floor ($\approx$ 7.5 Hz), and an aorta–heart meditation resonance (7 Hz). These empirical coincidences are *predictions*, not derivations. Five falsifiable tests are proposed.

**Keywords.** Golden ratio, Schumann resonance, alpha rhythm, DFT-8, bio-clocking, $\varphi$-ladder, formal verification, Lean 4

## 1 Introduction

Three empirical frequencies cluster near 7–8 Hz:

(i) The **Schumann fundamental** ($\approx$ 7.83 Hz): the lowest electromagnetic resonance of the Earth–ionosphere cavity (**??**).

(ii) The **alpha rhythm** (7.5–12 Hz): the dominant resting-state EEG oscillation (**??**).

(iii) **Bentov's aorta–heart resonance** ($\approx$ 7 Hz): a mechanical oscillation measured during meditation (**?**).

A body of empirical work has documented correlations between the Schumann resonance and human EEG activity (**???**). However, no existing framework derives a *specific* frequency window from first principles and shows that all three phenomena fall within it.

We present such a derivation within Recognition Science (RS), a cost-first formulation of physics derived from a single functional equation (**?**). The derivation yields the interval $(7.5, 9.3)$ Hz with zero adjustable parameters, conditional on one structural hypothesis (face-projection of the synchronization period) and one SI calibration ($\tau_0$).

**Provenance convention.** Claims carry one of three labels:

- **THEOREM**: forced by the RCL; machine-verified.
- **STRUCTURAL HYPOTHESIS**: follows from RS structure given a stated combinatorial choice; machine-verified conditionally.
- **EMPIRICAL PREDICTION**: involves contingent quantities; falsifiable.

# 2 Related Work

**?** provided quantitative evidence that Schumann resonance frequencies appear in human EEG power spectra, particularly in the theta and alpha bands. **?** demonstrated direct correlations between real-time Schumann resonance measurements and cerebral cortical activity. **?** proposed Schumann resonances as a biophysical mechanism linking solar/geomagnetic activity to human health.

These studies establish an empirical correlation but do not explain *why* the alpha band falls near the Schumann fundamental. The present work offers a candidate structural explanation: both frequencies are constrained by the same $\varphi$-ladder emanating from the 8-tick cycle.

# 3 The Forcing Chain

## 3.1 The primitive

**Definition 1** (Recognition Composition Law)**.** *For all $x, y > 0$:*

$$J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y). \tag{1}$$

With normalization $J(1) = 0$ and calibration $J''_{\log}(0) = 1$, the unique solution is

$$J(x) = \tfrac{1}{2}(x + x^{-1}) - 1 \tag{2}$$

(**?**). The following chain is forced:

## 3.2 Steps S1–S2: $\varphi$ and $D = 3$ (THEOREM)

**S1.** **$\varphi$ forced.** Self-similarity in the discrete ledger forces $x^2 = x + 1$; the unique positive root is $\varphi = (1 + \sqrt{5})/2$. *Lean*: `phi_sq_eq`.

**S2.** **$D = 3$ forced.** Topological linking requires $D = 3$; the minimal ledger-compatible period is $2^3 = 8$; the consciousness synchronization period is $S = \mathrm{lcm}(8, 45) = 360$. *Lean*: `sync_period_eq_360`, `dimension_forced`.

## 3.3 Step S3: Neural binding rung (STRUCTURAL HYPOTHESIS)

**S3.** **$N = 360/6 = 60$.** The $Q_3$ hypercube has $F = 2D = 6$ faces:

$$N = S/F = 360/6 = 60. \tag{3}$$

*Lean*: `neural_binding_rung_eq`.

**Remark 2** (Status of S3)**.** *The choice to divide by $F$ (faces) rather than $V = 8$ (vertices) or $E = 12$ (edges) is a **structural hypothesis**, not a theorem. Using vertices would give $N = 45$ and $f_{\text{cycle}} \in (75, 95)$ Hz; using edges would give $N = 30$ and $f_{\text{cycle}} \in (670, 880)$ Hz. Only face-projection ($N = 60$) places $f_{\text{cycle}}$ in the empirically observed gamma binding band (30–50 Hz).*

*We adopt this as a hypothesis with an explicit falsifier: if the gamma binding frequency is robustly measured outside* $(38, 45)$ *Hz, the face-projection hypothesis fails.*

*The physical motivation for face-projection is that each face of* $Q_3$ *corresponds to a two-dimensional "wallpaper" slice of the voxel lattice; the consciousness synchronization cycle projects onto each such slice, yielding a face-resolved period of* $S/F$*. A rigorous derivation from a variational principle remains open.*

### 3.4 Steps S4–S5: Bio-clocking and DFT-8 (THEOREM given S3)

**S4.** $f_{\text{cycle}} \in (38, 45)$ **Hz.** The Bio-Clocking Theorem gives $T = \tau_0 \cdot \varphi^N$ for the recognition cycle period at rung $N$. At $N = 60$:

$$T = \tau_0 \cdot \varphi^{60}, \qquad f_{\text{cycle}} = 1/T. \tag{4}$$

Interval arithmetic yields $3.44 \times 10^{12} < \varphi^{60} < 3.58 \times 10^{12}$ (proved in Lean via $\varphi^{60} = \varphi^{45} \times (\varphi^5)^3$), giving $f_{\text{cycle}} \in (38, 45)$ Hz.

*Lean*: `f_cycle_derived_bounds`.

*Note on* $\tau_0$: The value $\tau_0 = 7.30 \times 10^{-15}$ s is the single SI calibration in this derivation. It maps RS-native (dimensionless) timescales to seconds. $f_{\text{cycle}}$ depends linearly on $1/\tau_0$: a factor-of-2 change in $\tau_0$ would shift $f_{\text{cycle}}$ to $\sim$20 Hz, falsifying the prediction. Within RS, $\tau_0$ is derived from the coherence quantum $E_{\text{coh}} = \varphi^{-5}$ and the Planck constant via $\hbar = E_{\text{coh}} \cdot \tau_0$ (**?**); it is not a fit parameter but a derived SI conversion factor.

**S5. Mode 1** $\in (4.7, 5.7)$ **Hz.** The DFT-8 decomposes the 8-tick cycle into 7 non-DC modes at $f_k = k \cdot f_{\text{cycle}}/8$. Mode 1 satisfies $f_1 = f_{\text{cycle}}/8 \in (4.7, 5.7)$ Hz. *Lean*: `mode1_derived_bounds`.

### 3.5 Steps S6–S7: $\varphi$-harmonic (THEOREM)

**S6.** $\varphi$ **is the unique self-similar ratio.**

**Theorem 3.** *If* $r > 0$ *and* $r^2 = r + 1$*, then* $r = \varphi$*.*

*Proof.* $(r - \varphi)(r + \varphi) = r - \varphi$, so $(r - \varphi)(r + \varphi - 1) = 0$. Since $r > 0$ and $\varphi > 1$, $r + \varphi - 1 > 0$, forcing $r = \varphi$. *Lean*: `phi_unique_self_similar`. □

The cost function $J$ penalizes deviations from unity. Among all frequency ratios $r > 1$, those satisfying the self-similar recursion $r = 1+1/r$ (i.e., $r^2 = r+1$) are the scale-invariant fixed points of the $J$-cost landscape. Theorem **??** shows $\varphi$ is the only such ratio.

This does *not* mean $\varphi$-harmonics are the only resonances. Integer harmonics ($f_k = kf_1$) are the DFT modes themselves. The $\varphi$-harmonic is the unique *inter-mode* resonance forced by scale invariance; it fills the gap between adjacent DFT modes.

**S7.** $f_\varphi \in (7.5, 9.3)$ **Hz.** The first $\varphi$-harmonic of Mode 1 is $f_\varphi = f_1 \cdot \varphi$. From the bounds:

$$f_\varphi \in (4.7 \times 1.61,\ 5.7 \times 1.62) = (7.57,\ 9.23). \tag{5}$$

*Lean*: `f_phi_rung1_in_alpha`.

| **Forcing chain** |
|---|
| RCL $\xrightarrow{\text{T5}}$ $J$ unique $\xrightarrow{\text{T6}}$ $\varphi$ $\xrightarrow{\text{T8}}$ $D$=3, $S$=360 $\xrightarrow{\text{S3 (hyp.)}}$ $N$=60 $\xrightarrow{\text{S4}}$ $f_{\text{cycle}}\in(38, 45)$ $\xrightarrow{\text{S5}}$ $f_1\in(4.7, 5.7)$ $\xrightarrow{\text{S6}}$ $\varphi$ unique $\xrightarrow{\text{S7}}$ $f_\varphi\in(7.5, 9.3)$ Hz = **alpha band** |

Figure 1: Derivation chain. Step S3 is a structural hypothesis (face-projection); all other steps are theorems.

# 4 Empirical Predictions

The derived window makes three predictions. These are labeled **EMPIRICAL PREDICTION** because they involve contingent quantities.

## 4.1 Prediction 1: Schumann resonance

**Hypothesis 1** (Schumann in $\varphi$-window)**.** $7.5 < f_S < 9.3$ *Hz.*

*Status*: $f_S \approx 7.83$ Hz (**?**). Confirmed. The ratio $f_S/f_1 \approx 7.83/5 = 1.566$ is within 3.2% of $\varphi$. Ionospheric variability causes $f_S$ to fluctuate by $\pm 0.5$ Hz (**?**); the $(7.5, 9.3)$ window absorbs this.

## 4.2 Prediction 2: Alpha rhythm floor

**Hypothesis 2** (Alpha floor in $\varphi$-window)**.** *The lower bound of the alpha band ($\approx 7.5$ Hz) lies in* $(7.5, 9.3)$ *Hz.*

*Status*: Confirmed. RS predicts that the alpha floor is set by the $\varphi$-harmonic of Mode 1, offering a structural explanation for why alpha begins near 8 Hz rather than at an arbitrary frequency.

## 4.3 Prediction 3: Aorta–Heart Resonance

A mechanical resonance of the aorta–skull system during deep meditation has been measured at $\approx 7$ Hz.

**Hypothesis 3** (Meditation resonance in $[f_1, f_\varphi)$. $]4.7 < f_{\text{Meditation}} < 9.3$ *Hz.*

*Status*: $7.0 \in (4.7, 9.3)$. Confirmed, though the window is wide (4.6 Hz). This is the weakest of the three predictions and would benefit from tighter $\varphi^{60}$ bounds.

# 5 Convergence and Selection Effects

Table 1: Frequencies near the derived (7.5, 9.3) Hz window.

| Phenomenon | Frequency (Hz) | In window? | Source |
|---|---|---|---|
| DFT-8 Mode 1 (theta) | 4.7–5.7 | below | THEOREM |
| $\varphi$-harmonic of Mode 1 | 7.5–9.3 | = window | THEOREM |
| Schumann fundamental | 7.83 | yes | PREDICTION |
| Alpha rhythm floor | $\approx 7.5$ | boundary | PREDICTION |
| Aorta–heart resonance | 7.0 | yes (wide) | PREDICTION |
| Heart rate | $\sim 1$ | no | not predicted |
| Respiratory rate | $\sim 0.25$ | no | not predicted |
| Delta rhythm | 0.5–4 | no | not predicted |
| Beta rhythm | 13–30 | no | not predicted |

We emphasize that the present derivation predicts *only* the $\varphi$-harmonic of Mode 1. The framework does not claim to predict all biological frequencies. Heart rate ($\sim$ 1 Hz), respiratory rate ($\sim$ 0.25 Hz), and beta oscillations (13–30 Hz) are not addressed; their explanation would require identifying the relevant $\varphi$-ladder rungs and DFT modes, which is beyond the scope of this paper.

The apparent clustering of Schumann, alpha, and aorta–heart frequencies near 8 Hz is suggestive but not, by itself, statistically rigorous. The three frequencies are not fully independent: Schumann resonance is known to modulate EEG alpha power (**?**), and the meditation measurement was made during an alpha-dominant state. We do *not* claim a low $p$-value; we claim a structural derivation that produces the correct window.

# 6 The Neural Binding Rung

## 6.1 Construction

Two RS-derived quantities:

$$S = \mathrm{lcm}(8, 45) = 360 \qquad \text{(proved: } \texttt{sync_period_eq_360}\text{)} \tag{6}$$

$$F = 2D = 6 \qquad (\mathrm{Q}_3 \text{ faces}) \tag{7}$$

The **face-projection hypothesis** defines $N = S/F = 60$.

## 6.2 Alternative combinatorial choices

The $Q_3$ hypercube has $V = 8$ vertices, $E = 12$ edges, and $F = 6$ faces. Table **??** shows the consequences of each choice.

Table 2: Rung and cycle frequency for each cube-geometric divisor.

| Divisor | $N$ | $f_{\text{cycle}}$ (Hz) | Matches gamma? |
|---|---|---|---|
| $F = 6$ (faces) | 60 | 38–45 | yes |
| $V = 8$ (vertices) | 45 | $\sim 2000$ | no |
| $E = 12$ (edges) | 30 | $\sim 10^5$ | no |

Only face-projection yields a cycle frequency compatible with empirical gamma oscillations. This constitutes an argument from empirical adequacy, not a first-principles derivation. A rigorous proof that face-projection is the unique correct choice remains an open problem.

# 7 The $\varphi$-Ladder Frequency Bridge

**Theorem 4** ($\varphi$ uniqueness)**.** *If $r > 0$ and $r^2 = r + 1$, then $r = \varphi$.*

The DFT-8 modes at $f_k = k f_{\text{cycle}}/8$ are integer-spaced harmonics. Between adjacent modes $f_k$ and $f_{k+1}$, the scale-invariant cost landscape admits a $\varphi$-harmonic: $f_k \cdot \varphi$. This is the unique inter-mode resonance that satisfies the self-similar recursion $r = 1 + 1/r$ of the cost function.

For Mode 1 ($f_1 \approx 5$ Hz), the $\varphi$-harmonic at $\approx 8$ Hz falls between Mode 1 and Mode 2 ($\approx 10$ Hz). The alpha band, in this picture, occupies the spectral region between the theta fundamental and its first integer harmonic, anchored by the $\varphi$-ladder.

# 8 Machine-Checked Verification

Table 3: Lean 4 verification status. All modules compile with zero `sorry` and zero additional axioms.

| Module | Theorems | Sorry | Key result |
|---|---|---|---|
| `Cost.FrequencyLadder` | 7 | 0 | $\varphi$ unique self-similar |
| `Consc.NeuralBindingRung` | 12 | 0 | $N$=60, $f_{\text{cycle}} \in (38, 45)$ |
| `Consc.BodyCosmosResonance` | 10 | 0 | $f_\varphi \in (7.5, 9.3)$ |

# 9 Discussion

## 9.1 Forced versus predicted versus hypothesized

The derivation contains three epistemic layers:

1. **Theorems** (Steps S1–S2, S4–S7): forced by the RCL given the structural hypothesis of S3. These are machine-verified.
2. **Structural hypothesis** (Step S3): the face-projection $N = 360/6$. This is the single weakest link and the primary target for strengthening.
3. **Empirical predictions**: the Schumann, alpha, and Bentov frequencies falling in the window. These are falsifiable.

## 9.2 The role of $\tau_0$

The SI calibration $\tau_0 = 7.30 \times 10^{-15}$ s enters the derivation multiplicatively: $T = \tau_0 \cdot \varphi^{60}$. Changing $\tau_0$ by a factor $\lambda$ scales $f_{\text{cycle}}$ by $1/\lambda$. Within RS, $\tau_0$ is derived from the identity $\hbar = E_{\text{coh}} \cdot \tau_0$ with $E_{\text{coh}} = \varphi^{-5}$ and the SI-defined $\hbar$; it is a unit-conversion factor, not a fit parameter. Nonetheless, the Hz-valued prediction depends on this value, and any alternative determination of $\tau_0$ would shift the window proportionally.

## 9.3 Interpretation

The proximity of the Schumann resonance to the derived $\varphi$-window is *consistent with* the interpretation that the Earth's electromagnetic cavity resonates at a $\varphi$-harmonic of the DFT-8 fundamental. We stress that this is one possible interpretation of a frequency coincidence; it does not constitute proof that the Earth "is" a recognition system. The falsifiable content is the window prediction itself, not the ontological gloss.

## 9.4 Limitations

1. Step S3 (face-projection) is a hypothesis, not a theorem. Deriving it from a variational principle is the main open problem.
2. The $\varphi^{60}$ bounds are proved to $\sim 4\%$ precision. Tighter interval arithmetic would narrow the window and strengthen all three predictions.
3. The derivation addresses only the $\varphi$-harmonic of Mode 1. A complete bio-frequency theory from RS would need to account for delta, beta, and other rhythms via their respective $\varphi$-ladder rungs.
4. The aorta–heart resonance measurement is a single study from 1977. Modern replication with quantitative instrumentation would strengthen or refute Prediction 3.

## 10 Falsifiable Predictions

**F1.** If the Schumann fundamental is measured outside $(7.5, 9.3)$ Hz under standard ionospheric conditions, the window prediction fails.

**F2.** If the human alpha rhythm floor is robustly established below 7.0 Hz or above 10 Hz across large populations, the $\varphi$-harmonic explanation for the alpha floor fails.

**F3.** If meditation-induced body resonances are robustly measured outside $(4.7, 9.3)$ Hz, Prediction 3 fails.

**F4.** If the gamma binding frequency is robustly measured outside $(38, 45)$ Hz, the face-projection hypothesis ($N = 60$) fails.

**F5.** If EEG individual alpha frequency (IAF) distributions show no clustering near $\varphi \times f_1 \approx 8$ Hz across large datasets (**?**), the $\varphi$-ladder mechanism is unsupported.

## 11 Conclusion

Starting from the Recognition Composition Law, we have derived a frequency window $(7.5, 9.3)$ Hz using a chain of six theorems and one structural hypothesis (face-projection of the synchronization period). The Schumann resonance, the alpha rhythm floor, and the aorta–heart meditation resonance all fall in this window.

The derivation is machine-verified in Lean 4 with zero `sorry`. The single structural hypothesis ($N = 360/6$) is explicitly labeled and falsifiable via the gamma binding frequency. If strengthened to a theorem, the entire chain from the RCL to the alpha-band window would be parameter-free and fully forced.

## References


Balser, M. and Wagner, C. A. (1960). Observations of Earth–ionosphere cavity resonances. *Nature*, 188:638–641.

Berger, H. (1929). Über das Elektrenkephalogramm des Menschen. *Archiv für Psychiatrie und Nervenkrankheiten*, 87:527–570.

Cherry, N. (2002). Schumann resonances, a plausible biophysical mechanism for the human health effects of Solar/Geomagnetic activity. *Natural Hazards*, 26:279–331.

Grandy, T. H., Werkle-Bergner, M., Chicherio, C., Lövdén, M., Schmiedek, F., and Lindenberger, U. (2013). Individual alpha peak frequency is related to latent factors of general cognitive abilities. *NeuroImage*, 79:10–18.

Klimesch, W. (1999). EEG alpha and theta oscillations reflect cognitive and memory performance: a review and analysis. *Brain Research Reviews*, 29:169–195.

Persinger, M. A. (2014). Schumann resonance frequencies found in quantitative electroencephalographic activity: Real or spurious? *International Journal of Biometeorology*, 58:1483–1490.

Price, C., Williams, E., Elhalel, G., and Sentman, D. (2021). Natural ELF fields in the atmosphere and in living organisms. *International Journal of Environmental Research and Public Health*, 18:3165.

Saroka, K. S. and Persinger, M. A. (2011). Quantitative evidence for direct effects between Earth-ionosphere Schumann resonances and human cerebral cortical activity. *International Journal of Environmental Research and Public Health*, 8:1049–1064.

Schumann, W. O. (1952). Über die strahlungslosen Eigenschwingungen einer leitenden Kugel, die von einer Luftschicht und einer Ionosphärenhülle umgeben ist. *Zeitschrift für Naturforschung A*, 7:149–154.

Washburn, J. (2025). The algebra of reality: A Recognition Science derivation of physical law. *Axioms (MDPI)*, 15(2):90.

Washburn, J. (2026). Precision envelope targeting of neural oscillation bands. Preprint.